



Lecture 12: Introduction to Smart Grid

EEE 321: Power System I

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Presentation Outline

- ❑ Introduction to Smart Grid
- ❑ Expectations from Smart Grid
- ❑ Conventional Grid vs. Smart Grid
- ❑ Recent Challenges and Solutions
 - At consumer end- Demand Side Management
 - At generation end- Large Scale Renewable Integration

Introduction to Smart Grid

What is 'Smart grid'?

Today "Smart Grid" is used as a marketing term, rather than a technical definition. For this reason there is no well defined and commonly accepted scope of what it is and what it is not. The general understanding is that the Smart Grid is the concept of modernizing the electric Grid.



According to IEC

It is an electricity network that can intelligently integrate the actions of all users connected to it – generators, consumers and those that do both – in order to efficiently deliver sustainable, economic and secure electricity supplies.

According to DoE, US Government

“A smart grid uses digital technology to improve reliability, security, and efficiency (both economic and energy) of the electric system from large generation, through the delivery systems to electricity consumers and a growing number of distributed-generation and storage resources.”

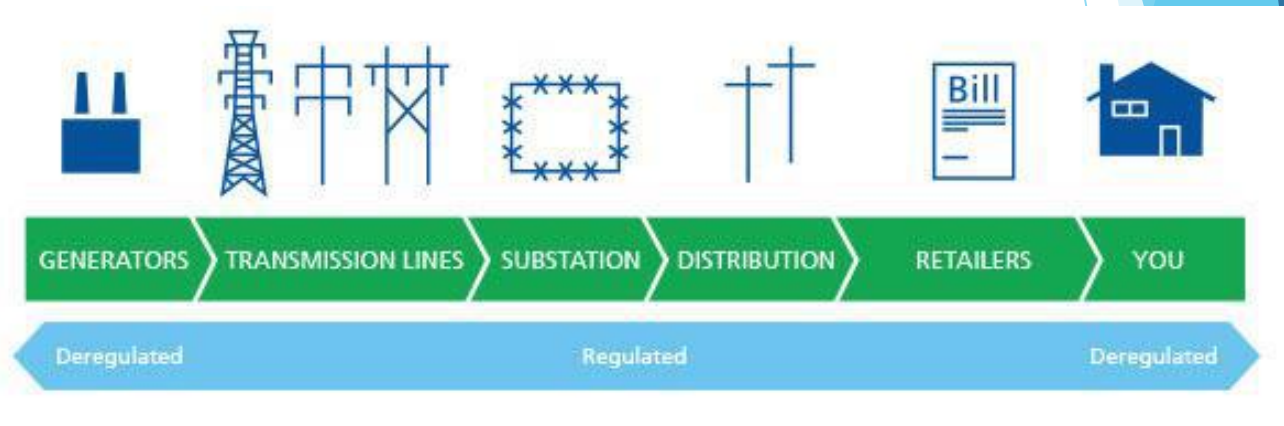
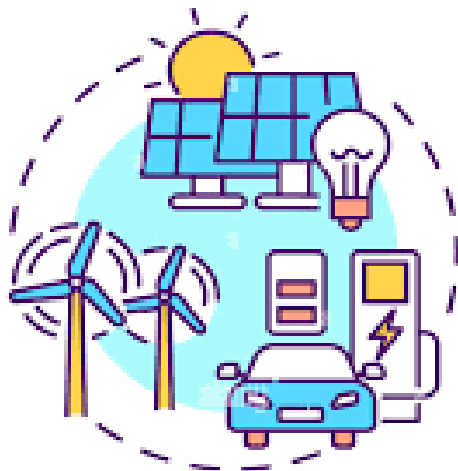
According to ABB

A smart grid is an evolved grid system that manages electricity demand in a sustainable, reliable and economic manner, built on advanced infrastructure and tuned to facilitate the integration of all involved.

Expectations from Smart Grid

A Smart Grid employs innovative products and services together with intelligent monitoring, control, communication, and self-healing technologies to:

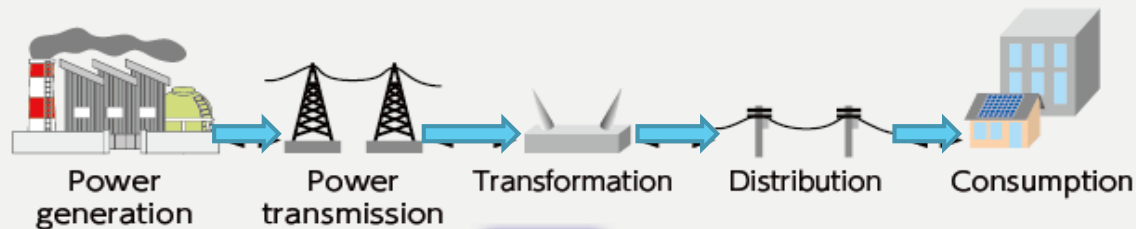
- ❑ facilitate the connection and operation of generators of all sizes and technologies;
- ❑ allow consumers to play a part in optimizing the operation of the system;
- ❑ provide consumers with greater information and choice of supply in a competitive market lowering the energy price;
- ❑ significantly reduce the environmental impact of the whole electricity supply system;
- ❑ deliver enhanced levels of reliability and security of supply.



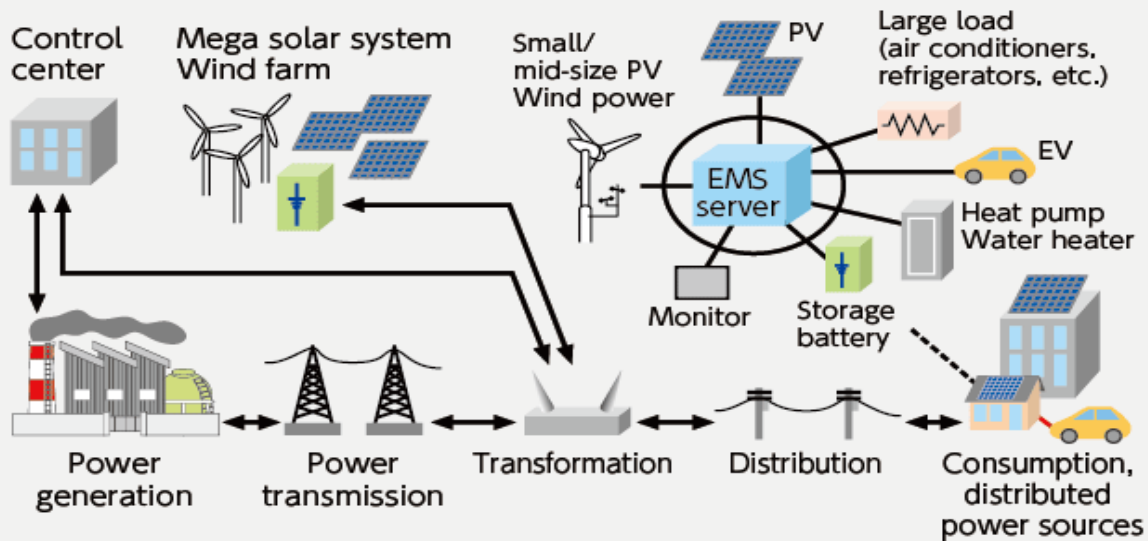
Comparison with conventional grid

Conceptual Diagram of Smart Grid

From grid to smart grid: Optimize power generation and consumption with IT-based efficient control

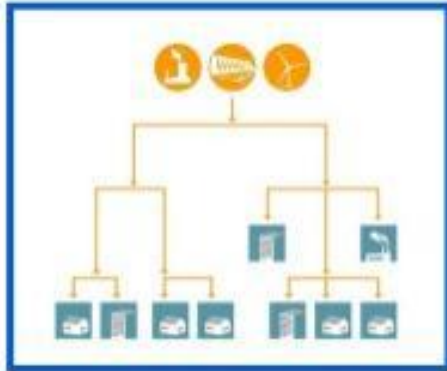


Smart Grid



Comparison with conventional grid

traditional grid



- Centralized power generation
- One-directional power flow
- Generation follows load
- Top-down operations planning
- Operation based on historical experience

smart grids

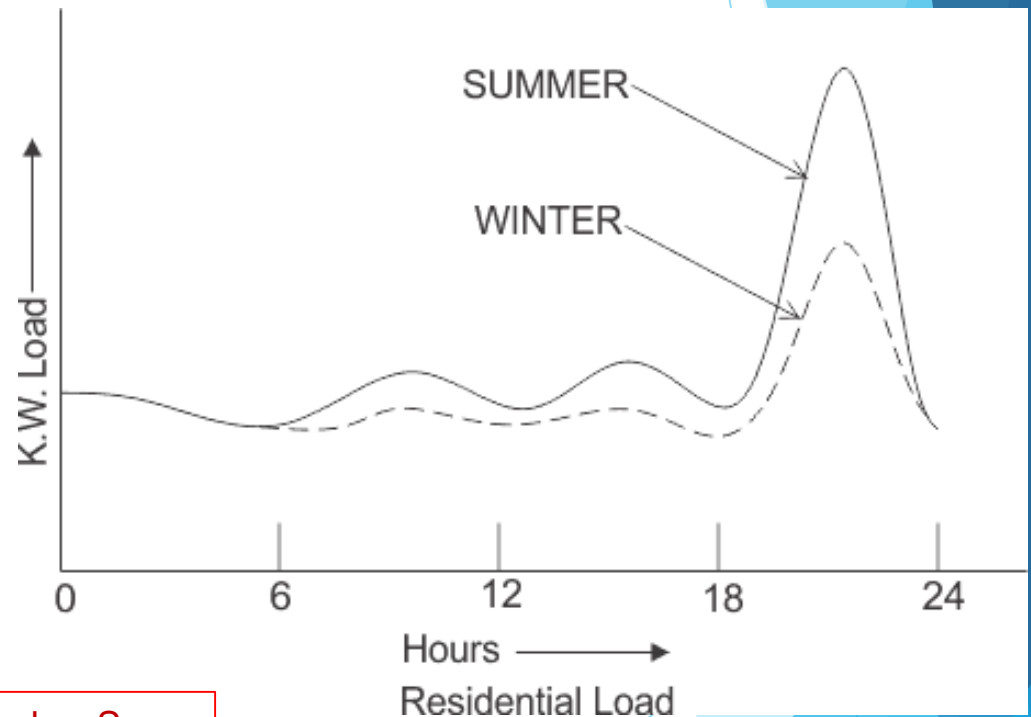


- Centralized and distributed power generation
- Volatile renewable power generation
- Multi-directional power flow
- Flexibility in demand – load follows generation
- Operation based on real-time data

Demand Side Management

In many countries, the power infrastructure is ageing and is being heavily used as demand for electricity rises. This overloading will worsen as large numbers of electric vehicles start consuming electricity from the electric power system. Obtaining planning permission for the installation of new power system equipment is becoming increasingly difficult. Therefore, demand-side programs have been introduced widely to make better use of the existing power supply infrastructure and to control the growth of demand.

The name Demand-Side Management (DSM) has been used since the 1970s for a systematic way of managing loads. Later on, Demand Response (DR), Demand-Side Response (DSR), Demand-Side Bidding (DSB) and Demand Bidding (DB) were used to describe a range of different demand side initiatives.



A household needs 5-7 kWh of electricity per day. So a car having 40 kWh capacity battery consumes energy equal to 6-7 households alone.

Demand Side Management (DSM)

- ❑ **Increasing Energy Demand and Consumption:** DSM helps manage and optimize energy use, reducing overall demand and improving energy efficiency. It also helps to **defer network investment**.
- ❑ **Focus on Energy Efficiency and Conservation:** Governments and organizations are placing greater emphasis on energy efficiency and conservation to reduce energy costs and environmental impact. These include subsidies, tax benefits, and mandates for energy-saving measures, which encourage the adoption of DSM solutions.
- ❑ **Advancements in Smart Grid Technology:** The development and deployment of smart grid technology facilitate more effective demand side management. Smart grids enable real-time monitoring and control of energy use, allowing for better demand response and energy management.
- ❑ **Growth in Renewable Energy Integration:** As the integration of renewable energy sources like solar and wind increases, DSM helps balance supply and demand by adjusting energy usage patterns. DSM solutions are crucial for managing the variability of renewable energy sources.
- ❑ **Rising Energy Costs:** Increasing energy prices drive both consumers and businesses to seek ways to manage and reduce their energy bills. DSM provides tools and strategies for optimizing energy consumption and lowering costs.

Demand-side resources such as flexible loads, distributed generation and storage can provide various services to the power system by modifying the load consumption patterns. Such services can include

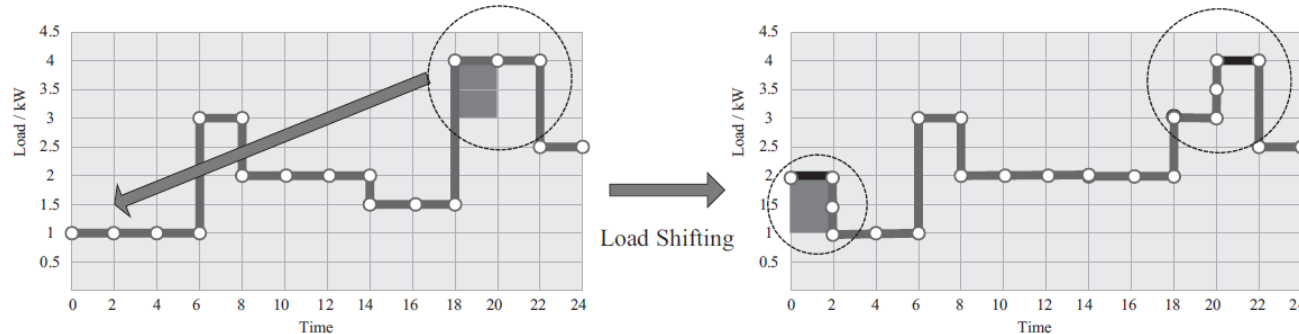
- ❑ Load shifting
- ❑ Valley filling
- ❑ Peak clipping
- ❑ Energy efficiency improvement

Simple daily domestic load profiles would be used to illustrate the function of each service, as shown in the next few slides.

Services provided by DSM

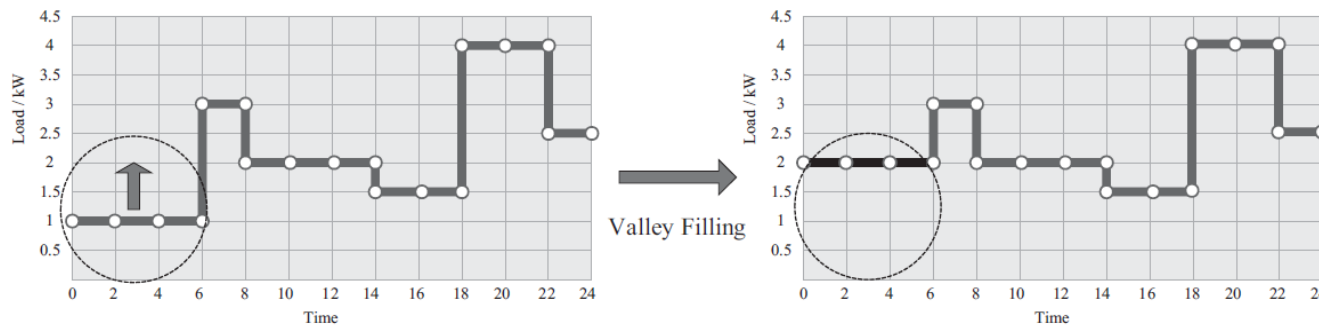
❑ Load shifting

Load shifting is the movement of load between times of day (from on-peak to off-peak) or seasons. In Figure below, a load such as a wet appliance (washing machine) that consumes 1 kW for 2 hours is shifted to off-peak time.



❑ Valley filling

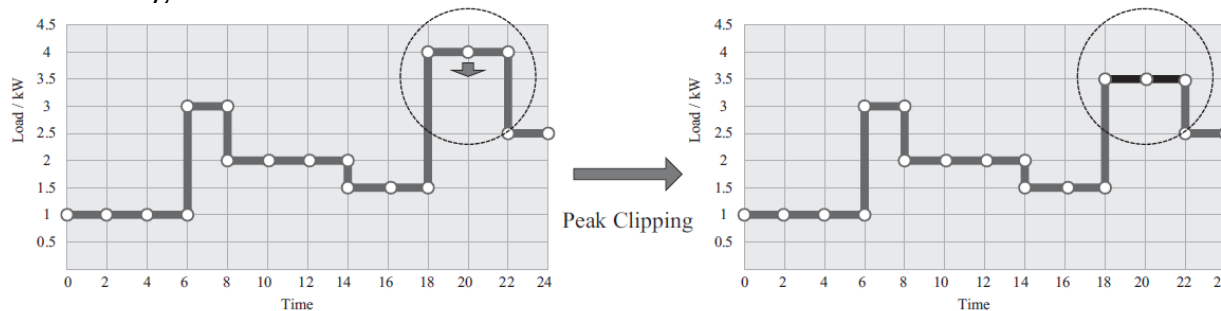
Figure below shows the main purpose of valley filling, which is to increase off-peak demand through storing energy, for example, in a battery of a plug-in electric vehicle or thermal storage in an electric storage heater. The main difference between valley filling and load shifting is that valley filling introduces new loads to off-peak time periods, but load shifting only shifts loads so the total energy consumption is unchanged.



Services provided by DSM

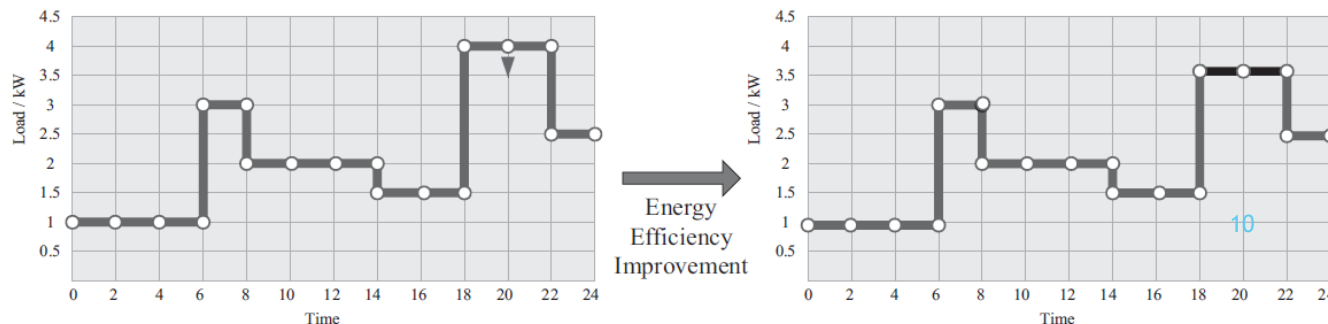
❑ Peak clipping

Peak clipping reduces the peak load demand, especially when demand approaches the thermal limits of feeders/transformers, or the supply limits of the whole system. Peak clipping is primarily done through direct load control of domestic appliances, for example, reducing thermostat setting of space heaters or control of electric water heaters or air-conditioning units. As peak clipping reduces the energy consumed by certain loads (in Figure below, 2 kWh of energy is reduced), often consumers have to sacrifice their comfort.



❑ Energy efficiency improvement

Energy efficiency programs are intended to reduce the overall use of energy. Approaches include offering incentives to adopt energy-efficient appliances, lighting, and other end-uses; or strategies that encourage more efficient electricity use, for example, the feedback of consumption and cost data to consumers, can lead to a reduction in total energy consumption. Figure below shows the reduction in energy demand when ten 60 W filament lamps (operating from 18.00 hrs to 22.00 hrs) are replaced by 20 W Compact fluorescent lamps.



Key Challenges with Large share of Renewables

The principal challenge for power system planning with high renewable energy penetration include managing the variability and intermittency of sources like wind and solar, requiring significant grid modernization and the integration of energy storage.

Variability and Intermittency:

Renewable energy sources like wind and solar are inherently variable and depend on weather conditions, leading to unpredictable and fluctuating power generation.

Grid Stability and Control:

The dynamic nature of renewables can challenge grid stability and frequency regulation, requiring advanced control systems and techniques to maintain a stable system.

Forecasting Uncertainty:

Accurately predicting the output of renewable energy sources is difficult, and errors in forecasting can lead to imbalances between supply and demand.

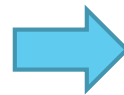
Protection system upgradation:

Protection systems must be upgraded to accommodate bidirectional power flows and the low, inverter-controlled fault currents characteristic of renewable sources, which can cause traditional relays to fail in detecting or accurately isolating faults.

Investment in Infrastructure Modernization:

Integrating large-scale renewables necessitates substantial investments in grid modernization, including the expansion of transmission and distribution networks to handle bidirectional power flows and increased generation as well as loads.

There is an urgent need for specialized planning techniques to ensure the entire power system can adapt to the fluctuations of ultra-high renewable energy penetration scenarios.



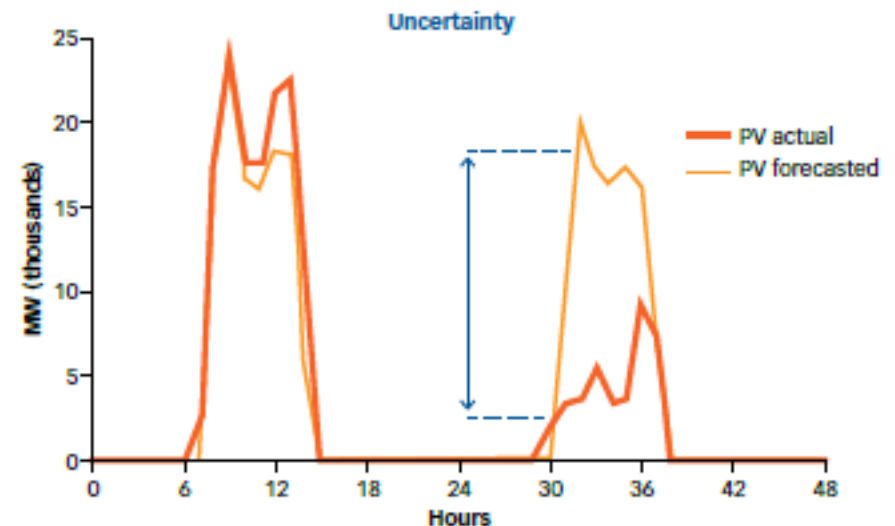
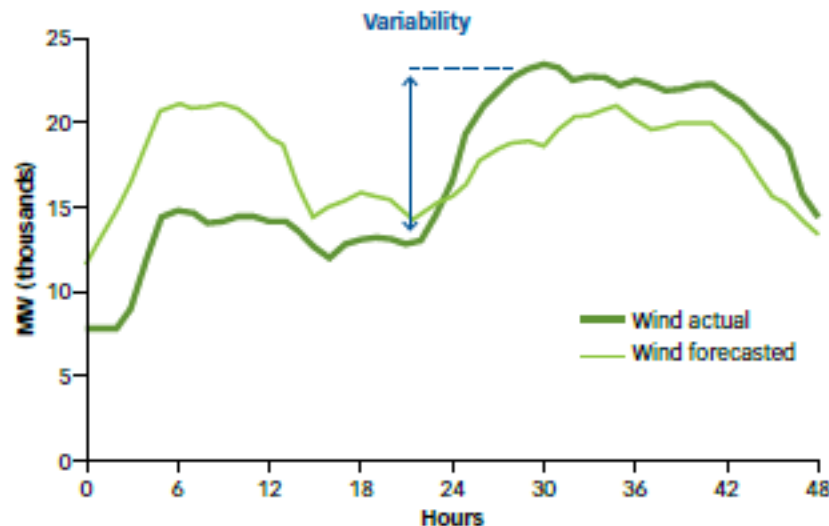
Key Challenges with Large share of Renewables

Variability, Intermittency, Forecasting uncertainty

Wind and solar resources produce electricity only intermittently—that is, when sun and wind are available. This means that, unlike fossil-fueled power, solar and wind power outputs at a specific location cannot be controlled at will. (They are “non-dispatchable.”)

Moreover, short-term variations in wind and solar resources cannot be perfectly predicted, even over the next hour or day (figure 1).

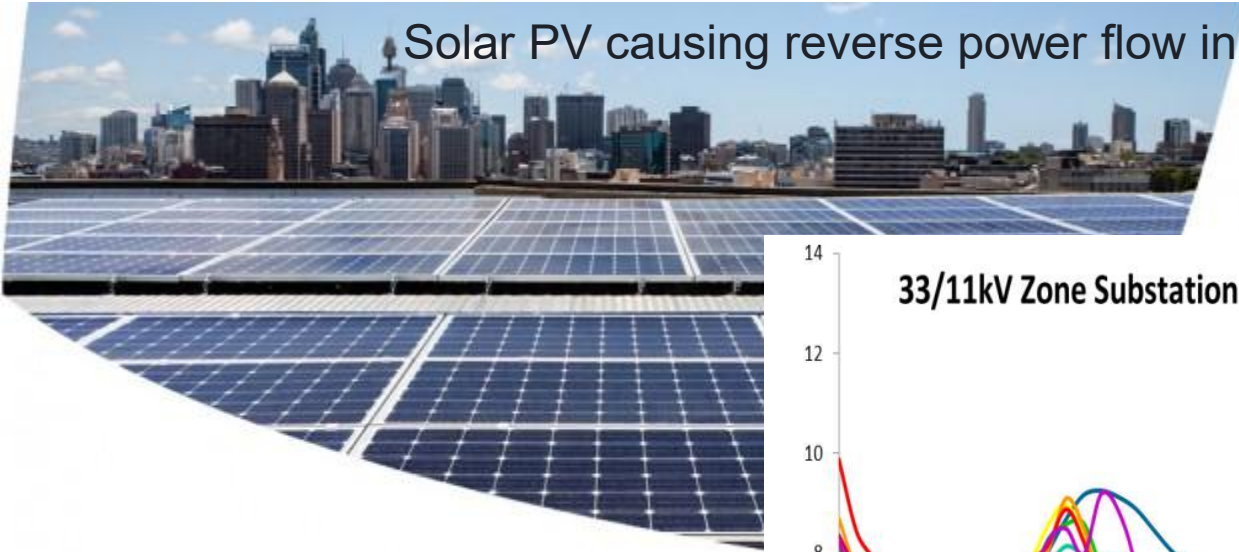
Figure 1. Variability of wind and solar output over a two-day period



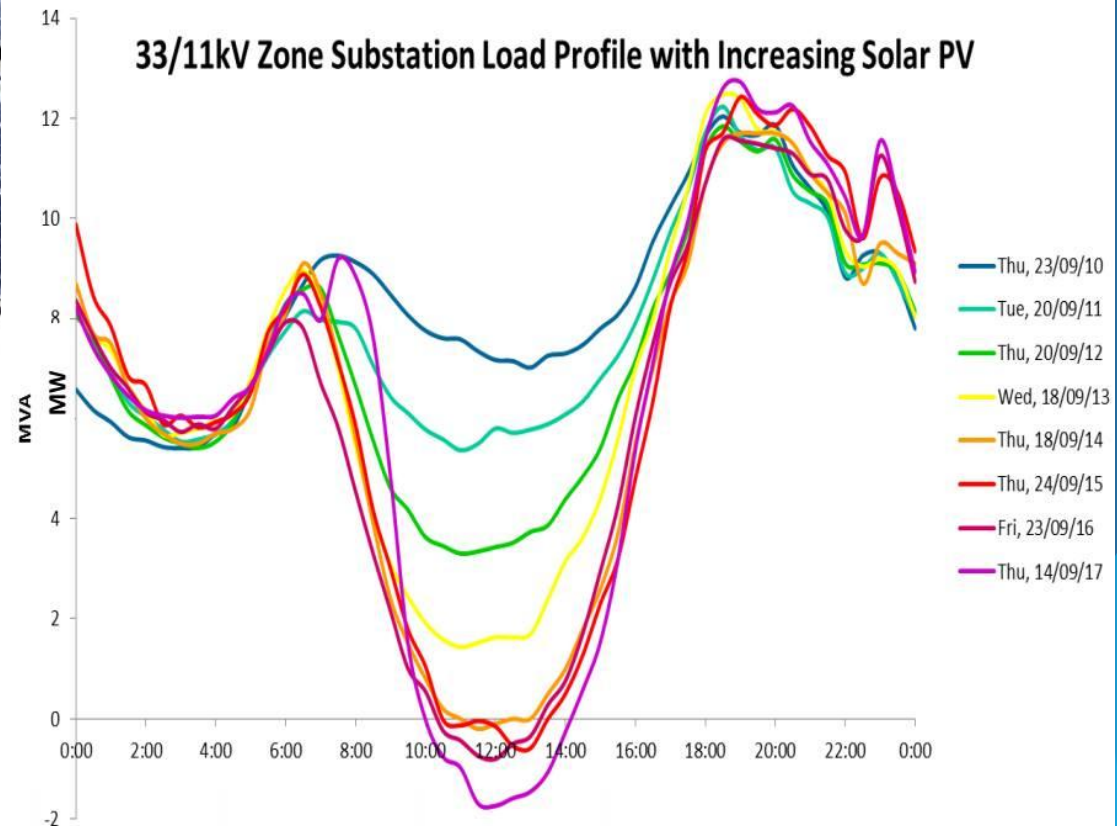
Key Challenges with Large share of Renewables

Protection System Upgradation

Solar PV causing reverse power flow in Queensland, Australia



Installed roof-top PV capacity: 2.016GW
(February, 2018)

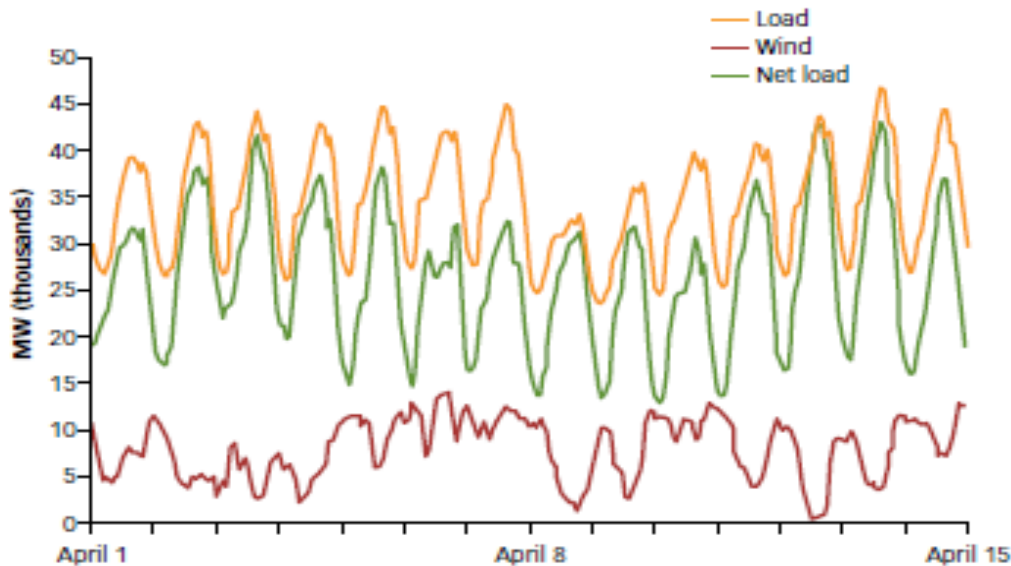


How to deal with Large share of Renewables

Adding variable renewable energy (VRE) based generation to a power system greatly increases uncertainty and variability, but solutions are available.

Integrating VRE sources into a power system creates something referred to as “net load” (net load = actual load – VRE production). As in a conventional system, the net load must be managed by grid operations, with the key difference that it is **more variable and less predictable** than the actual load (figure 3).

Figure 3. Movement of load, wind generation, and net load over a two-week period



Advanced Forecasting:

Utilizing advanced forecasting models and machine learning techniques can improve the accuracy of renewable energy generation predictions.

How to deal with Large share of Renewables

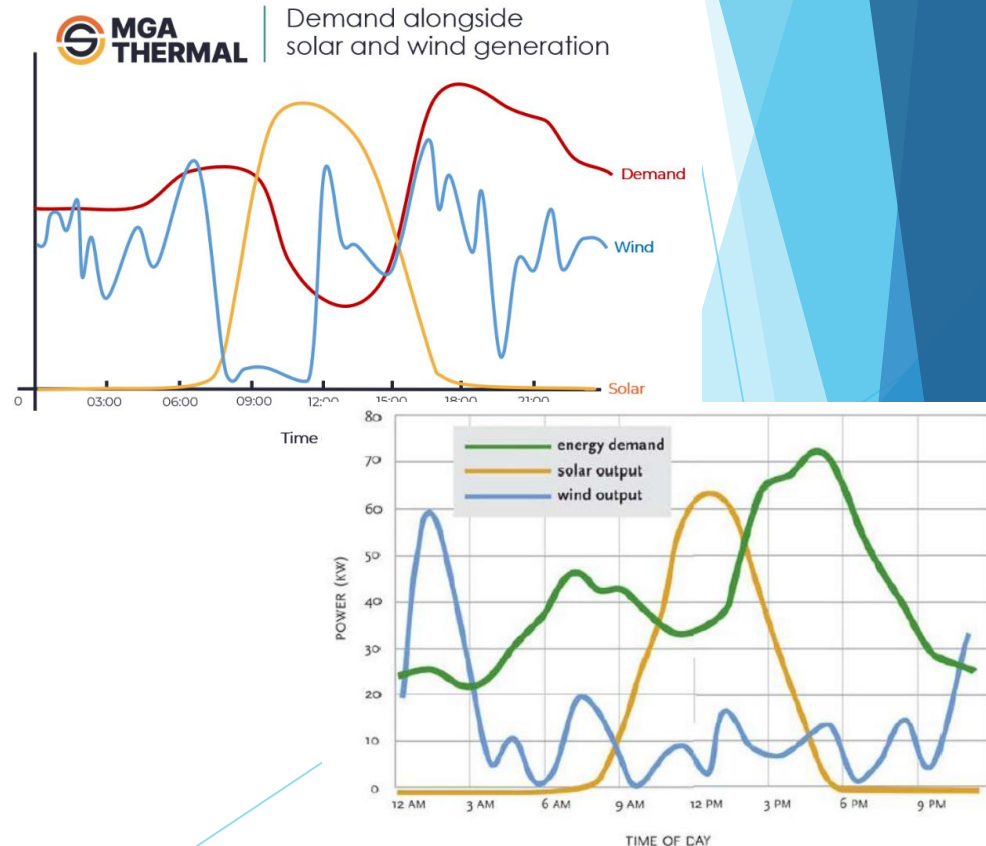
The ability of a power system to vary its output to **meet fluctuations in demand** depends on how fast its generation units can alter power output, a process known as ramping up (or down). Hydroelectric plants specially pumped hydro storage provides **quick ramping** compared to thermal plants. The other options include- battery energy storage systems (BESS), EV with V2G. Flexible natural gas combined cycle plants can also adjust their output rapidly.

Flexible Generation:

Deploying and integrating flexible conventional generation sources and other controllable resources can provide necessary backup and balancing services.

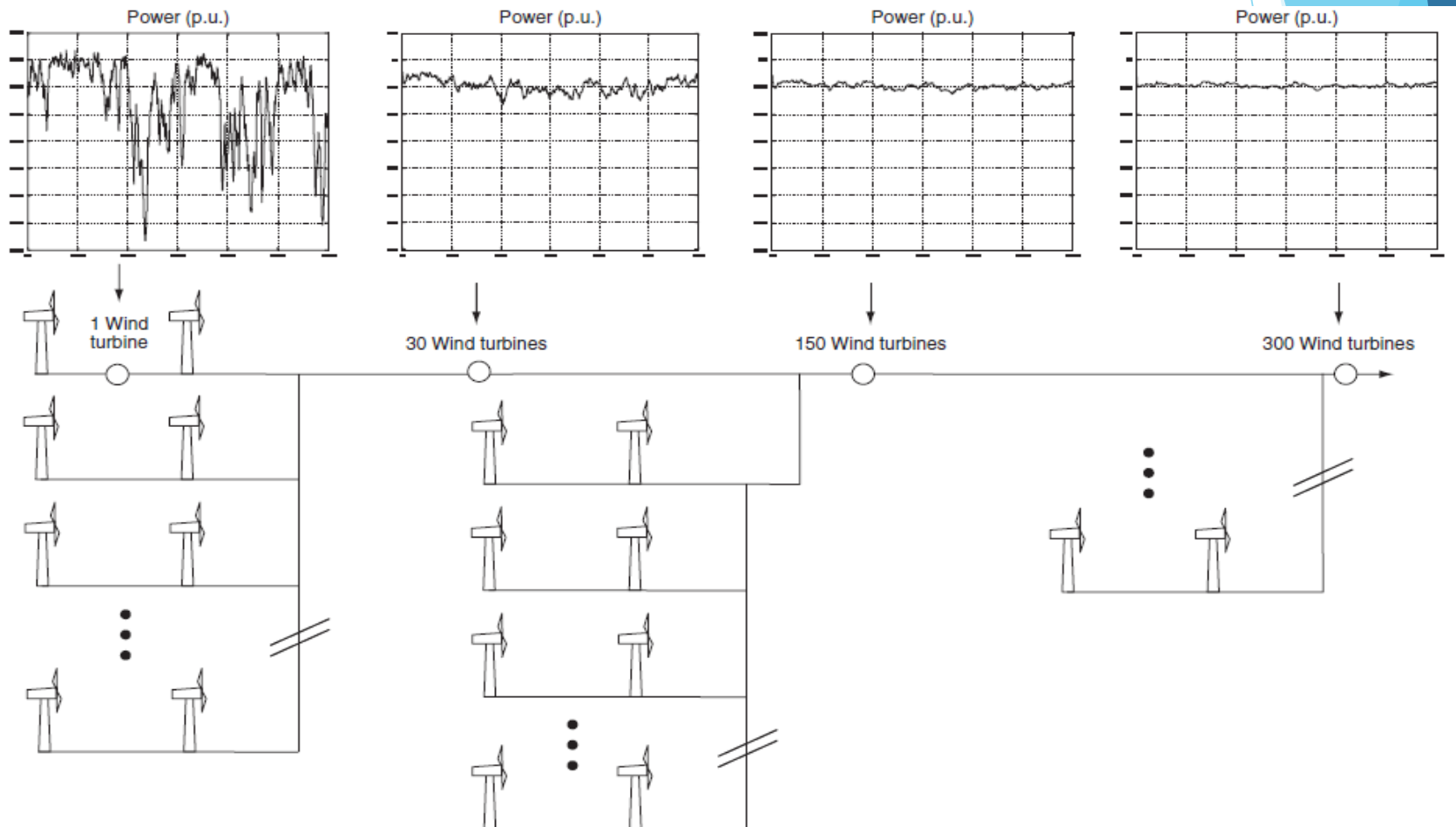
Combining resources to reduce variability:

Solar and wind resources are often negatively correlated: Solar power peaks in the summer, whereas wind tends to peak in the winter. On the other hand, solar power peaks during the day, while winds tend to be stronger in the afternoon and at night. These correlations make it possible to mix wind and solar resources to yield a combined power output that mimics the demand curve, a phenomenon called natural balancing. Aggregation of single resource helps to minimize intermittency issues.



How to deal with Large share of Renewables

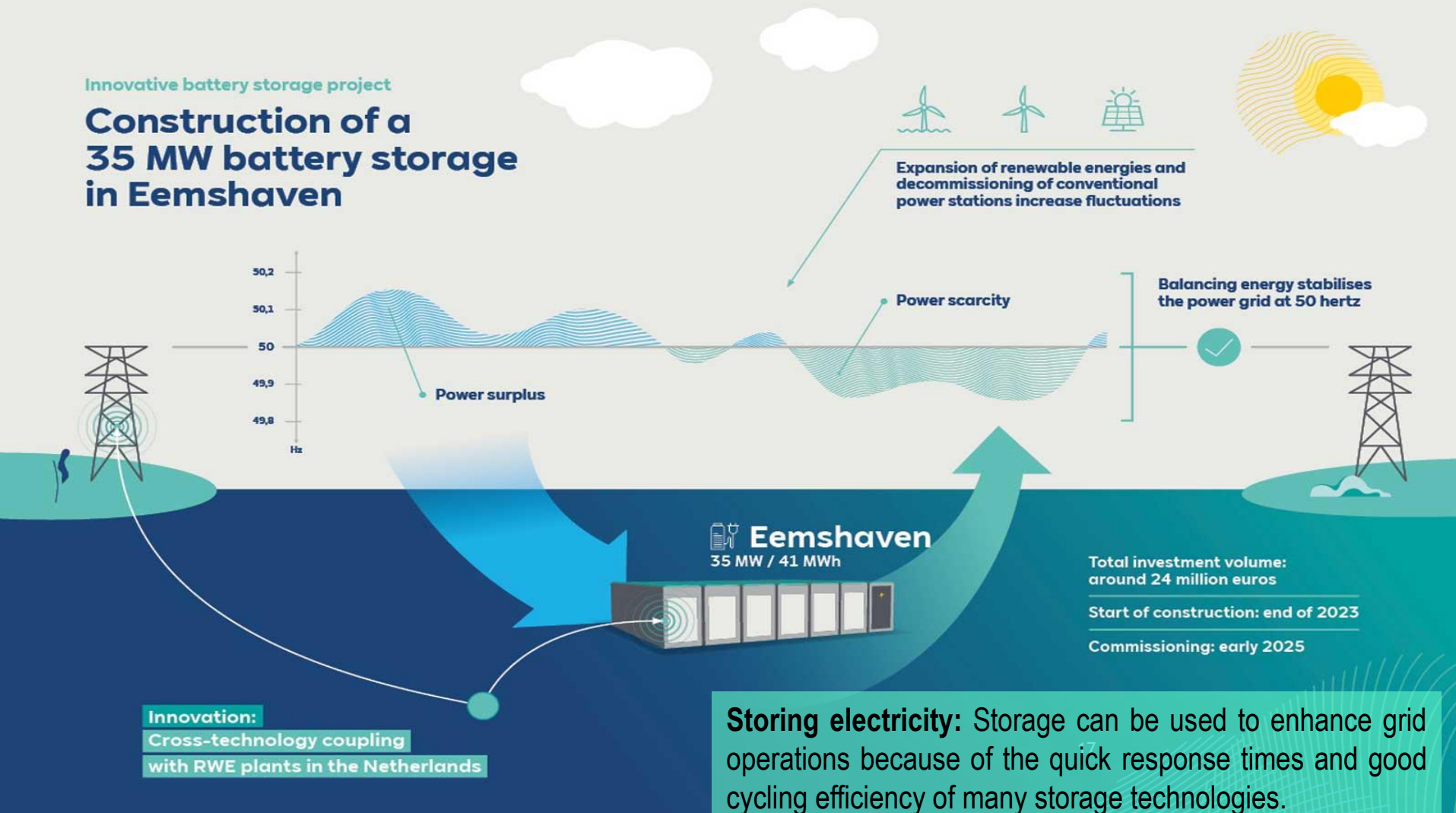
The aggregation of wind power provides an important positive effect on power system operation and power quality. Figure below shows the principal of aggregated wind power production.



How to deal with Large share of Renewables

Innovative battery storage project

Construction of a 35 MW battery storage in Eemshaven



References:

J. Ekanayake, Smart grid: Technology and Applications, Wiley-IEEE Press, 2012.



QUESTIONS AND DISCUSSIONS

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